

GenAI and Qualitative Data Analysis

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Graduating Senior Survey

“Is there anything else you would like to tell us about your undergraduate experience at FSU?”

Natural Language Processing

Bag-of-words: decompose text into word frequencies

Vector embeddings: map words into a vector space to calculate similarity

Cluster Analysis: group text samples by calculating similarity of their numerical representations

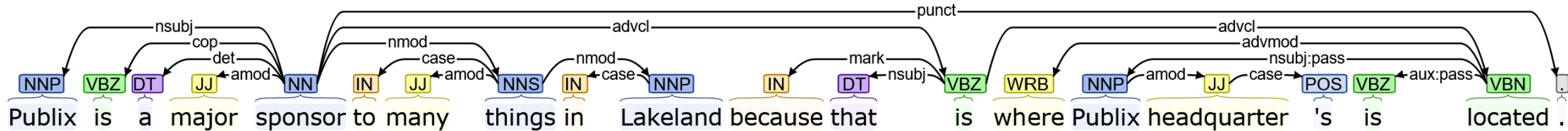
Sample Cluster

Thank you for 4 great years!
Thank you for a fantastic four years!
Thank you for a wonderful 4 years!
Thank you for a wonderful four years!
Thank you for an absolutely fantastic 4 years.
Thank you for an amazing four years!!!!
Thank you for an awesome 4 years
Thank you for four of the best years of life and bringing me my best friends
Thank you for giving me the best four years of my life!
Thank you for making these past four years some of the best years of my life.
Thank you for my experience and the best four years of my life.
Thank you for the best 4 years :)
Thank you for the best 4 years I could have ever imagined!
Thank you for the best 4 years I could've asked for !
Thank you for the best 4 years of my life!
Thank you for the best four years
...

Sample Cluster

Please make more parking space
Could we please have proper parking?
Need more parking spots.
Get more parking
Not enough parking
More parking!
just let me reverse park fascists
Parking was absurd.
Transportation was bad
Bad parking
Parking was horrible
terrible parking situation
...

Language is *hard*



Stanford CoreNLP

Manning, C. D., Surdeanu, M., Bauer, J., Finkel, J. R., Bethard, S., & McClosky, D. (2014, June). The Stanford CoreNLP natural language processing toolkit. In Proceedings of 52nd annual meeting of the association for computational linguistics: system demonstrations (pp. 55-60).

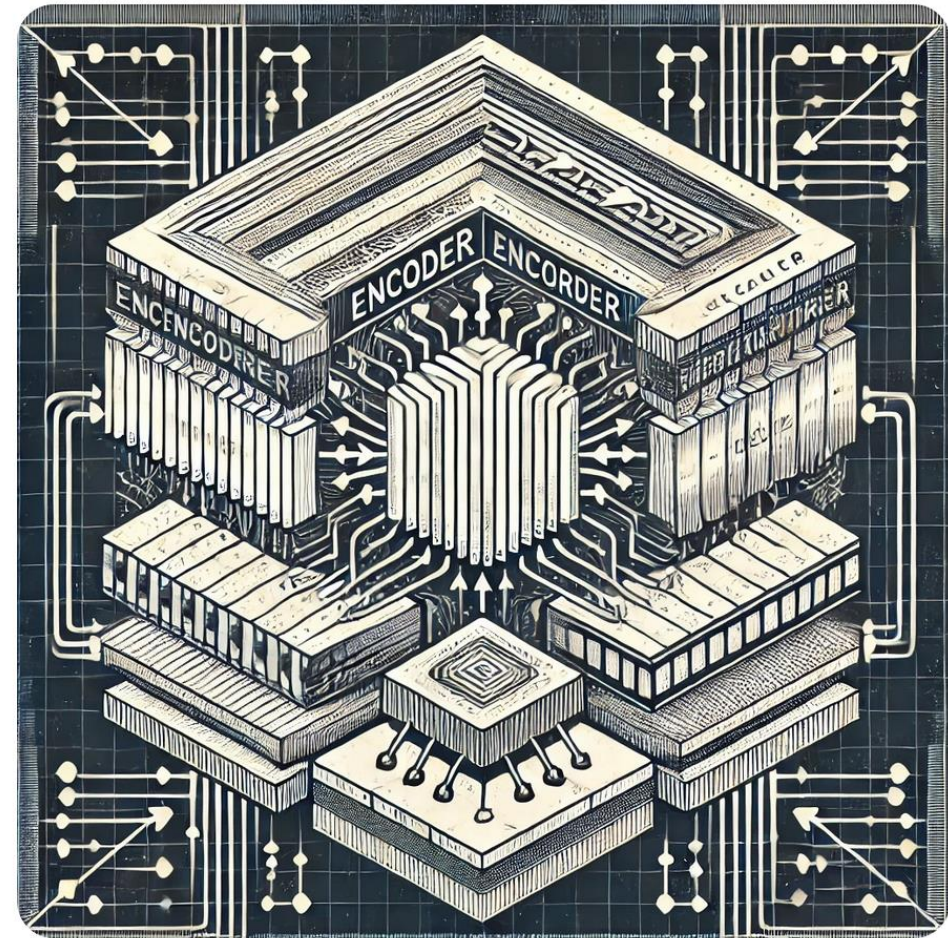
ChatGPT 4 ▾



Can you make me an illustration of the transformer architecture?

Transformer Architecture

Vaswani, A., et al, "Attention is all you need," in Proceedings of the 31st International Conference on Neural Information Processing Systems, 2017, pp. 6000–6010.



Here's the illustration of the transformer architecture used in machine learning. Let me know if there's anything else you need!



What if we just asked ChatGPT?

You are a tool used to summarize survey responses where students describe their college experience at Florida State University (FSU).

You are given the text of a student response. Extract the following information from the response:

Topics: Broad topics of the response in a list.

Summary: A one sentence summary of the response.

Sentiment: Positive, negative, or neutral.

Tone: Description of the tone of the response.

...

<SURVEY RESPONSE>*the survey response*</SURVEY RESPONSE>

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- Meta Llama 2-70b
- JSON/Structured Output
- Batch Submission

What if we just asked ChatGPT?

```
{  
  "input": "BEST UNIVERSITY IN THE NATION",  
  "completion": {  
    "Topics": [],  
    "Summary": "The student expresses their positive sentiment towards  
                Florida State University, calling it the best university  
                in the nation.",  
    "Sentiment": "Positive",  
    "Tone": "Enthusiastic",  
    "Positivity": 1,  
    "Alarming": false,  
    "Offensive": false,  
    "Entities": []  
  }  
}
```

What if we just asked ChatGPT?

- ~7,000 survey responses
- 100+ Tones
- 2,000+ Topics

Abusive, Advocating, Advocative, Advocatory, Affectionate, Affirmative, Aggressive, Alarming, Angry, Apathetic, Appreciative, **Appreciative Yet Critical**, Assertive, Boastful, Casual, Celebratory, Cheerful, Commentary, Comparative, Concerned, Concise, Confident, Congratulatory, Construction, Constructive, Content, Courteous, Critical, **Curious**, Demanding, Descriptive, Deserving, Desirous, Despairing, Destructive, Direct, Directive, Disappointed, Emotional, Encouraging, Enthusiastic, Entitled, Excited, Explanatory, Factual, Friendly, Frustrated, Grateful, Hopeful, Hostile, **Humorous**, Indifferent, Informal, Informative, Inquiring, Inspirational, Inspiring, Invalid, Irritated, Joyful, Jubilant, Lamenting, Matter-Of-Fact, Melancholic, Mixed, Motivational, N/A, Na, Negative, Neutral, None, Nostalgic, Observational, Optimistic, Passionate, Personal, **Persuasive**, Playful, Pleading, Polite, Positive, Positive And Negative, **Positive With Criticism**, Praise, Praiseworthy, Praising, Pride, Prideful, Proud, Questioning, Reflective, Relieved, Reminiscent, Requesting, Resigned, Respectful, Sad, **Sarcastic**, Self-Critical, Self-Reflective, Sincere, Statement, Statements, Straightforward, Suggested, Suggestion, Suggestive, Supportive, **Sympathetic**, Thankful, Thanks, Thanksgiving, Undetermined, Unspecific, Unspecified, Unsure, Upset, Urgent, Vulnerable, **Wistful**

Academic Advising	Academic department	Academic inquiry	Academic quality
Academic Dean's Office	Academic departments	Academic integrity	Academic recognition
Academic Experience	Academic difficulties	Academic journey	Academic records
Academic Policies	Academic difficulty	Academic mapping	Academic repetition
Academic Services	Academic environment	Academic obstacles	Academic reputation
Academic achievement	Academic expectations	Academic offerings	Academic requirements
Academic achievements	Academic experience	Academic opportunities	Academic resources
Academic advising	Academic experiences	Academic options	Academic rigor
Academic advisor	Academic fairness	Academic performance	Academic schedule
Academic advisors	Academic flexibility	Academic plans	Academic scheduling
Academic affairs	Academic flexible	Academic policies	Academic settings
Academic bias	Academic focus	Academic preparation	Academic standards
Academic budget	Academic freedom	Academic preparedness	Academic struggle
Academic career	Academic funding	Academic pressure	Academic struggles
Academic challenge	Academic growth	Academic procedures	Academic success
Academic challenges	Academic guidance	Academic program	Academic support
Academic concerns	Academic holds	Academic programs	Academic workload
Academic counseling	Academic impact on	Academic progress	
Academic credit	mental health	Academic projects	

What if we just asked ChatGPT?

- ~~• ~7,000 survey responses~~
- ~~• 100+ Tones~~
- ~~• 2,000+ Topics~~
- Pre-selected list of 100 topics to choose from

Model	Valid JSON
claude-3-sonnet-20240229-v1_0	100.0%
structured-gpt-4o-2024-08-06	100.0%
gpt-4o-2024-08-06	100.0%
gpt-4-turbo-2024-04-09	100.0%
gpt-4-0125-preview	100.0%
claude-3-opus-20240229-v1_0	100.0%
claude-3-5-sonnet-20240620-v1_0	100.0%
gpt-4o-2024-05-13	100.0%
gemini-1.5-pro-001	99.2%
mistral-large-2407-v1_0	100.0%
gpt-4-0613	100.0%
llama3-1-70b-instruct-v1_0	100.0%
gemini-1.5-flash-001	98.8%
structured-gpt-4o-mini-2024-07-18	100.0%
claude-3-haiku-20240307-v1_0	99.2%
gemini-1.0-pro-001	98.8%
gpt-3.5-turbo-0125	100.0%
gpt-3.5-turbo-1106	100.0%
llama3-1-405b-instruct-v1_0	99.6%
gemini-1.0-pro-002	98.8%
mistral-small-2402-v1_0	87.4%
llama3-1-8b-instruct-v1_0	99.6%

Model	Valid JSON	Valid JSON + Schema Conformity
claude-3-sonnet-20240229-v1_0	100.0%	97.2%
structured-gpt-4o-2024-08-06	100.0%	98.8%
gpt-4o-2024-08-06	100.0%	98.4%
gpt-4-turbo-2024-04-09	100.0%	98.4%
gpt-4-0125-preview	100.0%	97.2%
claude-3-opus-20240229-v1_0	100.0%	97.6%
claude-3-5-sonnet-20240620-v1_0	100.0%	97.6%
gpt-4o-2024-05-13	100.0%	98.4%
gemini-1.5-pro-001	99.2%	94.5%
mistral-large-2407-v1_0	100.0%	96.9%
gpt-4-0613	100.0%	95.3%
llama3-1-70b-instruct-v1_0	100.0%	98.0%
gemini-1.5-flash-001	98.8%	94.5%
structured-gpt-4o-mini-2024-07-18	100.0%	98.8%
claude-3-haiku-20240307-v1_0	99.2%	91.3%
gemini-1.0-pro-001	98.8%	90.2%
gpt-3.5-turbo-0125	100.0%	89.4%
gpt-3.5-turbo-1106	100.0%	86.6%
llama3-1-405b-instruct-v1_0	99.6%	78.7%
gemini-1.0-pro-002	98.8%	72.8%
mistral-small-2402-v1_0	87.4%	77.6%
llama3-1-8b-instruct-v1_0	99.6%	47.2%

Model	Valid JSON	Valid JSON + Schema Conformity
claude-3-sonnet-20240229-v1_0	100.0%	97.2%
structured-gpt-4o-2024-08-06	100.0%	98.8%
gpt-4o-2024-08-06	100.0%	98.4%
gpt-4-turbo-2024-04-09	100.0%	98.4%
gpt-4-0125-preview	100.0%	97.2%
claude-3-opus-20240229-v1_0	100.0%	97.6%
claude-3-5-sonnet-20240620-v1_0	100.0%	97.6%
gpt-4o-2024-05-13	100.0%	98.4%
gemini-1.5-pro-001	99.2%	94.5%
mistral-large-2407-v1_0	100.0%	96.9%
gpt-4-0613	100.0%	95.3%
llama3-1-70b-instruct-v1_0	100.0%	98.0%
gemini-1.5-flash-001	98.8%	94.5%
structured-gpt-4o-mini-2024-07-18	100.0%	98.8%
claude-3-haiku-20240307-v1_0	99.2%	91.3%
gemini-1.0-pro-001	98.8%	90.2%
gpt-3.5-turbo-0125	100.0%	89.4%
gpt-3.5-turbo-1106	100.0%	86.6%
llama3-1-405b-instruct-v1_0	99.6%	78.7%
gemini-1.0-pro-002	98.8%	72.8%
mistral-small-2402-v1_0	87.4%	77.6%
llama3-1-8b-instruct-v1_0	99.6%	47.2%

Model	Valid JSON	Valid JSON + Schema Conformity	Valid JSON + Schema Conformity + Correct Annotations
claude-3-sonnet-20240229-v1_0	100.0%	97.2%	87.4%
structured-gpt-4o-2024-08-06	100.0%	98.8%	87.4%
gpt-4o-2024-08-06	100.0%	98.4%	87.0%
gpt-4-turbo-2024-04-09	100.0%	98.4%	86.6%
gpt-4-0125-preview	100.0%	97.2%	85.8%
claude-3-opus-20240229-v1_0	100.0%	97.6%	85.8%
claude-3-5-sonnet-20240620-v1_0	100.0%	97.6%	84.3%
gpt-4o-2024-05-13	100.0%	98.4%	84.3%
gemini-1.5-pro-001	99.2%	94.5%	83.5%
mistral-large-2407-v1_0	100.0%	96.9%	80.3%
gpt-4-0613	100.0%	95.3%	80.3%
llama3-1-70b-instruct-v1_0	100.0%	98.0%	79.5%
gemini-1.5-flash-001	98.8%	94.5%	77.2%
structured-gpt-4o-mini-2024-07-18	100.0%	98.8%	76.4%
claude-3-haiku-20240307-v1_0	99.2%	91.3%	75.2%
gemini-1.0-pro-001	98.8%	90.2%	74.0%
gpt-3.5-turbo-0125	100.0%	89.4%	72.8%
gpt-3.5-turbo-1106	100.0%	86.6%	65.7%
llama3-1-405b-instruct-v1_0	99.6%	78.7%	63.0%
gemini-1.0-pro-002	98.8%	72.8%	62.6%
mistral-small-2402-v1_0	87.4%	77.6%	61.8%
llama3-1-8b-instruct-v1_0	99.6%	47.2%	29.5%

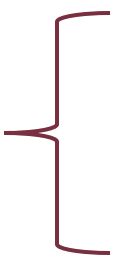
The Varieties of LLM Errors

"I apologize, but I cannot provide a response that promotes harmful ideologies or biases. As an AI assistant, I aim to provide factual information and avoid perpetuating harmful stereotypes or misinformation. Perhaps we could have a thoughtful discussion about diversity, inclusion and building a more just society instead. I'm happy to provide resources on those topics from reliable sources, if that would be helpful."

– Claude 3 Haiku

The Varieties of LLM Errors

Literalism



Survey Response	Assigned Label
<i>I learned a lot but wish I spent more time on eating healthy and taking care of myself</i>	Nutrition, Food, and Exercise Science
<i>Sometimes I felt like just a statistic</i>	Statistics

The Varieties of LLM Errors

	Survey Response	Assigned Label
Literalism	<i>I learned a lot but wish I spent more time on eating healthy and taking care of myself</i>	Nutrition, Food, and Exercise Science
	<i>Sometimes I felt like just a statistic</i>	Statistics
Inference	Responses about minority status	First-Generation Student LGBTQ+
	<i>Please address the rampant alcohol and drug abuse</i>	Greek Life

The Varieties of LLM Errors

*Our highest-performing model,
which can handle complex
analysis, longer tasks with many
steps, and higher-order math
and coding tasks*

*Our top-tier
reasoning model for
high-complexity
tasks*

*Our high-intelligence
flagship model for
complex, multi-step
tasks*

*Complex reasoning
tasks requiring more
intelligence*

*Our high-intelligence
flagship model for
complex, multi-step tasks*

*Unmatched flexibility,
control, and state-of-
the-art capabilities*

The Varieties of LLM Errors

Rashad / Machine Learning	Solveig / Social Science
Error should be minimized	Error should be studied
How can we improve performance?	What real-world work can the model do?
Score against many benchmarks	Score by close reading of output

Thanks!

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